

Nova Safety and Cybersecurity Assurance Report

Report scope

Requirements evaluated: 12

Domains: Functional safety, Cybersecurity

Methods: bench test, environmental exposure, inspection, log review

Result vocabulary: PASS, FAIL, NOT TESTED, INCONCLUSIVE

1. Executed results

Each disposition applies only to the criterion stated in the same record.

REQ-NVA-FSA-001 Watchdog reaction	INCONCLUSIVE - Work was attempted for REQ-NVA-FSA-001, but incomplete scope or calibration prevents deciding: Control task misses two consecutive execution windows. INCONCLUSIVE - Work was attempted for REQ-NVA-FSA-001, but incomplete scope or calibration prevents deciding: Independent watchdog resets the controller within 50 ms.	Bench / inspection record Reviewed by DV team
REQ-NVA-FSA-002 Sensor plausibility	PASS - Executed for REQ-NVA-FSA-002; the recorded result confirms: Redundant temperature-channel disagreement exceeds 8 deg C. PASS - Executed for REQ-NVA-FSA-002; the recorded result confirms: Diagnostic fault is set within 200 ms.	Bench / inspection record Reviewed by DV team
REQ-NVA-FSA-003 Torque inhibition	PASS - Executed for REQ-NVA-FSA-003; the recorded result confirms: Charge connector is latched. IN PROGRESS - Partial execution exists for REQ-NVA-FSA-003, but verification is not complete: Commanded propulsion torque remains zero. NOT TESTED - No test or inspection was performed for REQ-NVA-FSA-003: Vehicle displacement does not exceed 150 mm.	Bench / inspection record Reviewed by DV team
REQ-NVA-FSA-004 Safe-state persistence	PASS - Executed for REQ-NVA-FSA-004; the recorded result confirms: Crash latch is set. PASS - Executed for REQ-NVA-FSA-004; the recorded result confirms: Both contactors remain open. PASS - Executed for REQ-NVA-FSA-004; the recorded result confirms: Automatic restart is suppressed. FAIL - Executed for REQ-NVA-FSA-004; the observed result does not satisfy: Crash latch persists across a 12 V power cycle.	Bench / inspection record Reviewed by DV team

2. Executed results

Each disposition applies only to the criterion stated in the same record.

REQ-NVA-FSA-005 Fault logging	NOT TESTED - No test or inspection was performed for REQ-NVA-FSA-005: The reaction is an ASIL-D safety reaction. NOT TESTED - No test or inspection was performed for REQ-NVA-FSA-005: Safety reaction stores its fault code. NOT TESTED - No test or inspection was performed for REQ-NVA-FSA-005: First-failure timestamp resolution is 10 ms or better. NOT TESTED - No test or inspection was performed for REQ-NVA-FSA-005: At least 2 seconds of pre-trigger sensor history is stored.	Bench / inspection record Reviewed by DV team
REQ-NVA-CYS-001 Secure boot	PASS - Executed for REQ-NVA-CYS-001; the recorded result confirms: Signature verification occurs at every startup. PASS - Executed for REQ-NVA-CYS-001; the recorded result confirms: Application code executes only after successful verification.	Bench / inspection record Reviewed by DV team
REQ-NVA-CYS-002 Authentication lockout	PASS - Executed for REQ-NVA-CYS-002; the recorded result confirms: Five consecutive invalid authentication attempts occur. IN PROGRESS - Partial execution exists for REQ-NVA-CYS-002, but verification is not complete: Security access is locked for at least 15 minutes.	Bench / inspection record Reviewed by DV team
REQ-NVA-CYS-003 Update freshness	PASS - Executed for REQ-NVA-CYS-003; the recorded result confirms: Candidate image has a valid signature but a lower security version. FAIL - Executed for REQ-NVA-CYS-003; the observed result does not satisfy: Bootloader rejects the rollback image.	Bench / inspection record Reviewed by DV team

3. Executed results

Each disposition applies only to the criterion stated in the same record.

REQ-NVA-CYS-004 SecOC verification	NOT TESTED - No test or inspection was performed for REQ-NVA-CYS-004: The CAN message is protected. NOT TESTED - No test or inspection was performed for REQ-NVA-CYS-004: Freshness value is verified. NOT TESTED - No test or inspection was performed for REQ-NVA-CYS-004: Message authentication is verified before payload release. NOT TESTED - No test or inspection was performed for REQ-NVA-CYS-004: Failed verification increments the intrusion counter.	Bench / inspection record Reviewed by DV team
REQ-NVA-CYS-005 Key handling	INCONCLUSIVE - Work was attempted for REQ-NVA-CYS-005, but incomplete scope or calibration prevents deciding: Keys are production private keys. INCONCLUSIVE - Work was attempted for REQ-NVA-CYS-005, but incomplete scope or calibration prevents deciding: Production private keys are generated inside the HSM. INCONCLUSIVE - Work was attempted for REQ-NVA-CYS-005, but incomplete scope or calibration prevents deciding: Private keys are never exported in plaintext. INCONCLUSIVE - Work was attempted for REQ-NVA-CYS-005, but incomplete scope or calibration prevents deciding: Keys are erased after authorized ownership transfer.	Bench / inspection record Reviewed by DV team

4. Image-based inspection result

Visual Inspection Record

REQ-NVA-FSA-006 | Redundant shutdown path

INCONCLUSIVE

INCONCLUSIVE - Work was attempted for REQ-NVA-FSA-006, but incomplete scope or calibration prevents deciding: Primary safety MCU de-energizes the coils within 25 ms.

INCONCLUSIVE

INCONCLUSIVE - Work was attempted for REQ-NVA-FSA-006, but incomplete scope or calibration prevents deciding: Independent comparator de-energizes the coils within 25 ms.

REQ-NVA-CYS-006 | Debug-port state

PASS

the recorded result confirms: Controller is in production mode.

PASS

the recorded result confirms: External debug connector is physically unpopulated.

PASS

the recorded result confirms: Debug fuse reports disabled.

PASS

the recorded result confirms: Secure-state indicator appears green.

Image record NOVA-DV-VIS | Reviewer: QA-17 | Controlled copy

Figure 4-1. Recorded inspections for REQ-NVA-FSA-006, REQ-NVA-CYS-006; interpretation requires the raster content.